

PERSONALIZED LEARNING PATH RECOMMENDER

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HOUSE TARGARYEN

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1. The Learning Discovery Problem

Online learning platforms contain thousands of courses. The challenge is not only finding a course, but finding a sensible sequence that matches the learner's goal.

01

Too Many Choices

Large catalogs make it difficult to identify where to start.

02

Unstructured Learning

Learners may collect courses without a clear progression.

03

Need for Personalization

A recommendation should reflect the learner's intent and context.

2. What Our System Delivers

The project combines machine-learning retrieval with conversational AI to turn a learner's review or question into actionable recommendations.



Conversational AI Guidance

Ask a learning question or describe a target skill using a selected AI provider.



Personalized Top 10

Retrieve relevant courses using TF-IDF and NearestNeighbors similarity.



Structured Roadmaps

Break complex topics into three achievable milestones.



Interactive Dashboard

Chat history, progress checklists, provider switching, and API-key management.



Fallback Mode

Structured recommendations remain available without an external API key.

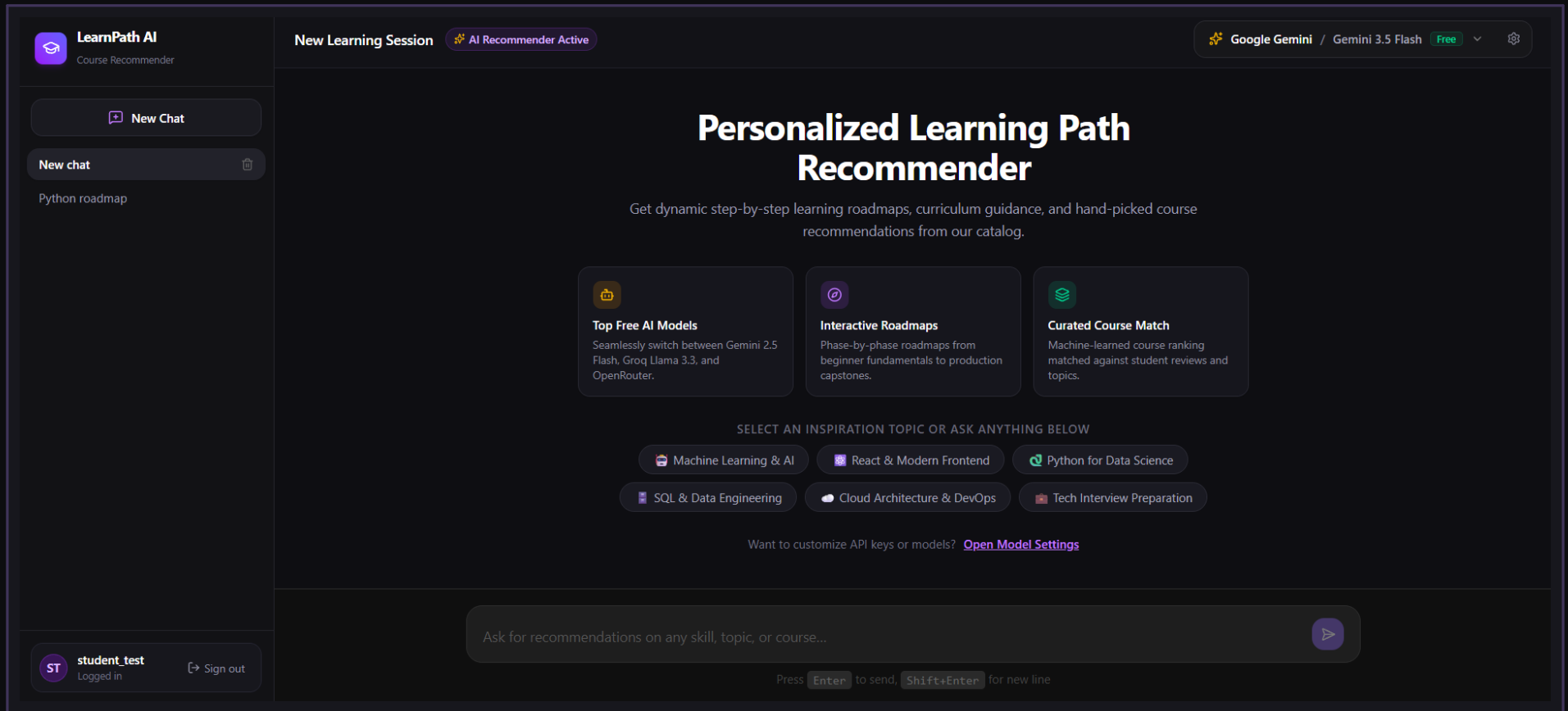


Course-Aware Ranking

Candidate recommendations are re-ranked before the final Top 10.

3. Product Interface

The interface follows a dark, modern dashboard style with purple accents, compact cards, model switching, topic starters, and a conversational input area.



4. Machine Learning Pipeline

STEP	PIPELINE STAGE	DETAIL
01	Training Reviews	train.csv
02	Text Preprocessing	clean_text
03	TF-IDF	100k features • 1-3 n-grams
04	Candidate Retrieval	NearestNeighbors • cosine distance • k=50
05	Classification	LogisticRegression • target course category
06	Re-ranking	Course-aware Top 10
07	Serving	FastAPI /ai/chat and /recommend

5. System Architecture

USER	→	REACT 19 + TYPESCRIPT	→	FASTAPI
Goal / review / question		Chat • recommendations • roadmap • progress		Auth • AI providers • recommendation API
		↓		↓
		TF-IDF + NearestNeighbors + LogisticRegression		Gemini • Groq • OpenRouter



6. Technology Stack

AREA	TECHNOLOGIES
Frontend	React 19 • TypeScript • Vite 8 • Tailwind CSS v4 • Lucide React
Backend	Python 3.10+ • FastAPI • Uvicorn • HTTPX • Pandas • NumPy
Machine Learning	Scikit-learn • TF-IDF • NearestNeighbors • LogisticRegression • LabelEncoder
AI Providers	Google Gemini • Groq Cloud • OpenRouter
Authentication	Passlib/bcrypt • python-jose • JWT

7. Example User Journey

STEP	ACTION	OUTPUT
01	Enter a goal	"I want to learn Deep Learning with PyTorch."
02	Understand context	The query is processed and enriched with recommendations.
03	Rank courses	TF-IDF + nearest-neighbor retrieval identifies candidates.
04	Generate guidance	The selected AI provider creates a structured response.
05	Build roadmap	Three phases define foundations, progression, and milestones.
06	Track progress	Checklists, history, and suggested prompts support continuation.

8. Evaluation & Deployment

- Recall@10: measures the proportion of relevant course reviews surfaced within the Top 10 recommendations for each query.
- Training: the pipeline fits the TF-IDF vectorizer, NearestNeighbors index, and LogisticRegression classifier on the training dataset.
- Batch prediction: test reviews produce recommendations in outputs/submission.csv.
- Deployment: FastAPI exposes authentication, health, AI, and recommendation endpoints while the React/Vite frontend provides the UI.

Core API Endpoints

ENDPOINT	PURPOSE
/health	Service status and model-loaded state
/ai/providers	Available LLM providers and model configuration
/ai/chat	Conversational AI enriched with recommendations and roadmaps
/recommend	Pure ML recommendation endpoint

THANK YOU

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AI

Conversational learning guidance.



ML

Course relevance and ranking.



ROADMAPS

Structured milestones and progress.